map-charting-student-math-misunderstandings

9th Place Solution

Rank	9
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Topic ID	612105
Version	Original

Source: https://www.kaggle.com/competitions/map-charting-student-math-misunderstandings/discussion/612105
Retrieved with Kaggle CLI on 2026-10-21.

Many thanks to the organizers and Kaggle staff, and congratulations to all the prize winners.
This competition taught me a lot — and I'm happy to have earned my first gold medal!

Overview

- **Method**
  - Used 32B Causal LM with vLLM for inference: 5-fold × 2 models = 10 total.
- **Scores**
  - Public LB: **0.950**, Private LB: **0.948**
- **Models**
  - `Qwen/Qwen2.5-32B-Instruct`
  - `deepseek-ai/DeepSeek-R1-Distill-Qwen-32B`
- **Inference time**
  - 460 minutes

Dataset

- Competition train data
  - Split into 5 folds with StratifiedKFold based on QuestionId, Category, Misconception.

Training

- **Reference implementation:**
  https://www.kaggle.com/code/sinchir0/lb-0-942-train-fullft-qwen3-8b-by-sfttrainer
- The 65 labels were converted into special tokens.
- Used **SFTTrainer + AutoModelForCausalLM** with **QLoRA** for fine-tuning.
- **Hyperparameters**
  - **Qwen/Qwen2.5-32B-Instruct**
    - `epoch=2, lr=1e-4, per_device_batch_size=8, gradient_accumulation_steps=2`
    - `r=32, α=64, lora_dropout=0.01, target_modules=["q_proj","k_proj","v_proj","o_proj","gate_proj","up_proj","down_proj"], modules_to_save=["lm_head"], trainable_token_indices={"embed_tokens": target_label_ids}`
  - **DeepSeek-R1-Distill-Qwen-32B**
    - `epoch=2, lr=8e-5, per_device_batch_size=8, gradient_accumulation_steps=2`

        - `r=64, α=128, lora_dropout=0.01, target_modules=["q_proj","k_proj","v_proj","o_proj","gate_proj","up_proj","down_proj"], modules_to_save=["lm_head"], trainable_token_indices={"embed_tokens": target_label_ids}`

- **Environment**
    - Google Colab (A100 80GB)
- **Training time**
    - 220 minutes / 1 fold

Preprocessing

- Fixed known annotation errors and unified duplicate Misconception labels.
- The unified labels were redistributed in post-processing based on the corresponding QuestionID.

```
() -> pd.DataFrame:
   false_to_true_ids = [, , , , , ]
   df[] = np.where(
       df[].isin(false_to_true_ids),
       df[]..replace(, ),
       df[]
   )

   true_to_false_ids = [, , , , , , , , , , , ]
   df[] = np.where(
       df[].isin(true_to_false_ids),
       df[]..replace(, ),
       df[]
   )
    df

() -> pd.DataFrame:
   df[] = df[].replace({: })
    df
```

Prompt Optimization (for Accuracy Improvement)

- Simplifying the prompt slightly improved CV scores.
- It also helped reduce training and inference time.

```
prompt_format =
```

Inference

- Used **vLLM + AWQ** for inference.

- Retrieved logits for the added special tokens, and averaged the scores (simple addition mean) across all models.

Results (MAP@3)

Model / Setting	Valid	Public LB	Private LB
Qwen2.5-32B-Instruct (fold0)	0.950182	0.946	0.944
Qwen2.5-32B-Instruct (5-fold)		0.950	0.946
DeepSeek-R1-Distill-Qwen-32B (fold0)	0.947956	0.948	0.945
DeepSeek-R1-Distill-Qwen-32B (5-fold)		0.950	0.946
5-fold × 2 ensemble		0.950	**0.948**

Attempts That Didn’t Improve Results

- **Reranker (pairwise / pointwise)**
  Implemented for top1/top2 reordering, but performance did not surpass the base results.
- **70B / 72B models**
  CV was better than 32B, but inference time (~150 min) was too long for final pipeline.
- **Non-Qwen2.5 models**
  Tried Llama 3.3-70B, gpt-oss-20b, Mistral, Hermes, Gemma 3, Qwen 3, etc.
  None outperformed Qwen 2.5 models in our situation.
- **Soft labels**
  Re-trained Qwen-32B using logits produced by another Qwen-32B model trained on the same dataset.
  No improvement was observed on the public leaderboard, but the private leaderboard score increased slightly (+0.001).
- **LoRA Merge**
  Merging LoRA adapters trained on five folds resulted in lower performance compared to using a single model.
- **Training only on [Category: Misconception] and excluding True/False samples**
  Excluding True/False samples and training only on the Misconception category did not outperform the standard category classification model.

References

https://huggingface.co/Qwen/Qwen2.5-32B-Instruct
https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Qwen-32B

https://www.kaggle.com/code/sinchir0/lb-0-942-train-fullft-qwen3-8b-by-sfttrainer
https://huggingface.co/docs/peft/main/en/developer_guides/lora#efficiently-train-tokens-alongside-lora